

Algebra 8 Placement Test — ANSWER KEY

Sienna Ash · 100 points total

Grader use only. Worked solutions shown for every item. Bold red = expected answer.

Section 1 — Evaluating Expressions & Function Notation

14 pts

Q1.

2 pts

Evaluate $3x - 5$ when $x = 4$.

$$3(4) - 5 = 12 - 5 = \mathbf{7}$$

Q2.

4 pts

Evaluate $2a^2 - 3b$ when $a = -2$ and $b = 5$.

$$2(-2)^2 - 3(5) = 2(4) - 15 = 8 - 15 = \mathbf{-7}$$

Q3.

4 pts

Evaluate $\frac{5x - y}{2}$ when $x = 3$ and $y = -1$.

$$\frac{5(3) - (-1)}{2} = \frac{15 + 1}{2} = \frac{16}{2} = \mathbf{8}$$

Q4.

4 pts

Let $f(x) = 3x - 1$.

(a) Find $f(-2)$.

$$f(-2) = 3(-2) - 1 = -6 - 1 = \mathbf{-7}$$

(b) Find the value of x where $f(x) = 11$.

$$3x - 1 = 11 \Rightarrow 3x = 12 \Rightarrow x = \mathbf{4}$$

Section 2 — Solving Equations

30 pts

Q5.

4 pts

$$2x + 9 = 21$$

$$2x = 12 \Rightarrow x = \mathbf{6}$$

Q6.

4 pts

$$3(x - 4) = 15$$

$$3x - 12 = 15 \Rightarrow 3x = 27 \Rightarrow x = \mathbf{9}$$

Q7.

6 pts

$$4x - 7 = x + 8$$

$$4x - x = 8 + 7 \Rightarrow 3x = 15 \Rightarrow x = \mathbf{5}$$

Q8.

6 pts

$$2(3x + 1) = 4x - 6$$

$$6x + 2 = 4x - 6 \Rightarrow 2x = -8 \Rightarrow x = \mathbf{-4}$$

Q9.

4 pts

$$\frac{2}{3}x - 4 = 6$$

$$\frac{2}{3}x = 10 \Rightarrow x = 10 \cdot \frac{3}{2} \Rightarrow x = \mathbf{15}$$

Q10.

3 pts

$$5x + 2 = 5x - 3$$

Subtract $5x$: $2 = -3$ — false. **No solution.**

Q11.

3 pts

$$4(x - 1) = 4x - 4$$

Distribute: $4x - 4 = 4x - 4 \Rightarrow 0 = 0$ — always true. **Infinitely many solutions.**

Section 3 — Proportional Relationships

18 pts

Q12.

6 pts

Table: (0,3), (2,7), (4,11), (6,15)

(a) **Not proportional** — when $x = 0$, $y = 3 \neq 0$. A proportional relationship passes through the origin.

(b) y-intercept = **3**

(c) Slope: $\Delta y / \Delta x = 4/2 = 2$. Equation: $y = 2x + 3$

Q13.

6 pts

Line through (0, -2) and (4, 6).

(a) **Not proportional** — y-intercept is -2 , not 0 .

(b) $m = \frac{6 - (-2)}{4 - 0} = \frac{8}{4} = \mathbf{2}$

(c) $y = 2x - 2$

Q14.

6 pts

Parking garage: \ entry + \ /hour.

(a) $C = 3h + 5$

(b) Slope = **cost increases by \ per additional hour** (the per-hour rate).

(c) **Not proportional** — the \ flat fee means $C = 5$ when $h = 0$, so the y-intercept is 5 , not 0 .

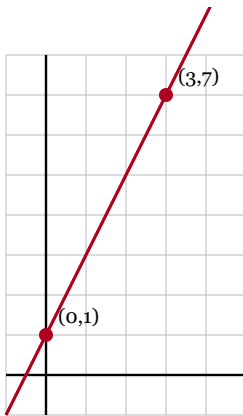
Section 4 — Identifying Slope

20 pts

Q15.

4 pts

Line through $(0, 1)$ and $(3, 7)$.

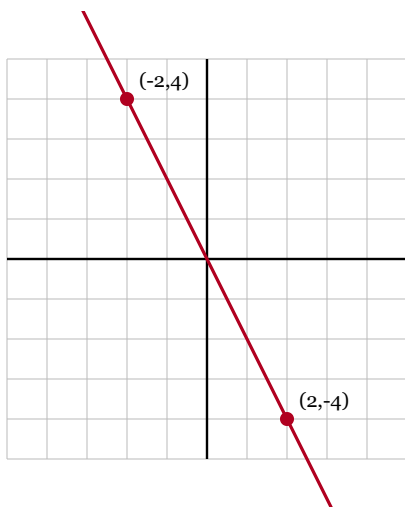


$$m = \frac{7 - 1}{3 - 0} = \frac{6}{3} = \mathbf{2}$$

Q16.

4 pts

Line through $(-2, 4)$ and $(2, -4)$.



$$m = \frac{-4 - 4}{2 - (-2)} = \frac{-8}{4} = -2$$

Q17.

6 pts

Line through $(-1, 5)$ and $(3, -3)$.

$$m = \frac{-3 - 5}{3 - (-1)} = \frac{-8}{4} = -2$$

Q18.

3 pts

Slope of $y = -4x + 7$?

Coefficient of x : $m = -4$

Q19.

3 pts

y-intercept of $y = 3x - 5$?

Constant term: $b = -5$

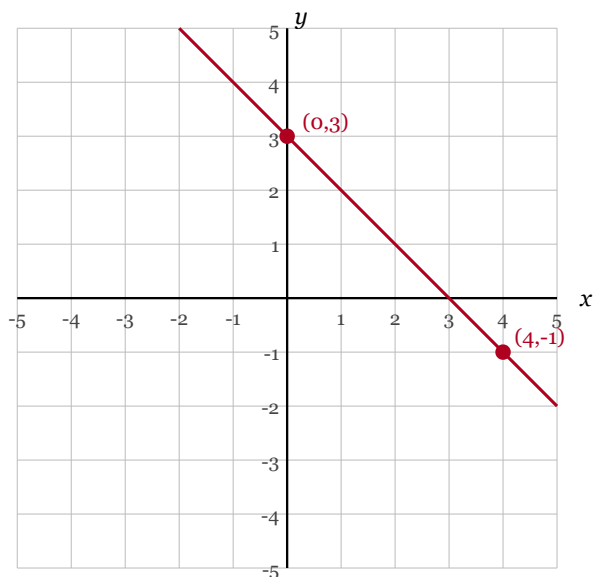
Section 5 — Graphing Lines

18 pts

Q20.

6 pts

Line through $(0, 3)$ and $(4, -1)$.



$$(a) m = \frac{-1 - 3}{4 - 0} = \frac{-4}{4} = -1$$

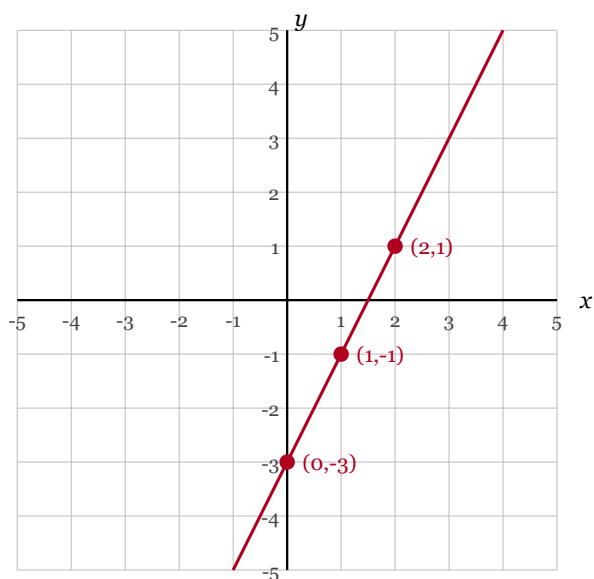
$$(b) \text{y-intercept} = 3$$

$$(c) y = -x + 3$$

Q21.

6 pts

Graph $y = 2x - 3$.

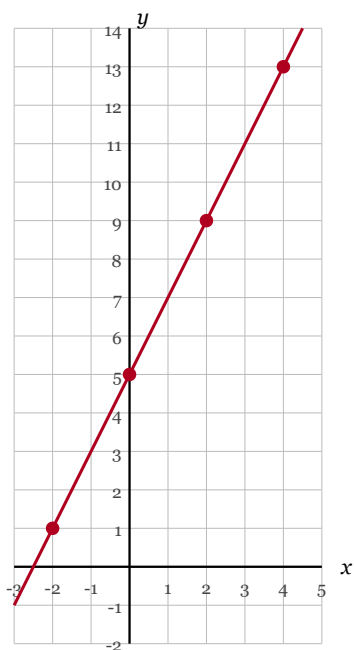


y-intercept: **(0, -3)**. Slope $2 = \frac{2}{1}$: from y-int, go up 2 right 1 to **(1, -1)**, then again to **(2, 1)**. Line continues both directions.

Q22.

6 pts

Table: (-2,1), (0,5), (2,9), (4,13).



Slope: $m = \frac{5 - 1}{0 - (-2)} = \frac{4}{2} = 2$. y-intercept: 5. Equation: $y = 2x + 5$.

Grading Summary

Section	Topic	Max	Earned	%
S1	Evaluating Expressions & Function Notation	14		
S2	Solving Equations	30		
S3	Proportional Relationships	18		
S4	Identifying Slope	20		
S5	Graphing Lines	18		
Total		100		

Placement thresholds:

- **85–100:** Ready for Algebra 8
- **70–84:** Likely ready — review gap areas first
- **55–69:** Borderline — pre-algebra reinforcement recommended
- **Below 55:** Not yet — continue 7th-grade math fundamentals

Concern flags (regardless of total score):

- S2 below 21/30 → equation-solving gap; focus on multi-step, variables-on-both-sides, and solution-type recognition.
- Q10 and Q11 both wrong → doesn't yet recognize no-solution / infinitely-many-solutions cases. Spend a week on this before Algebra 8.
- S5 below 13/18 → graphing gap; pre-algebra reinforcement on plotting and slope-intercept form recommended before Algebra 8.